Aiko — Due-Diligence Dossier

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> ***Confidential.*** *Prepared for investor due diligence. Not for distribution. This dossier describes proprietary architecture; please do not share it.*

What Aiko is, in one paragraph

Aiko is an agent that lives on your laptop and does the things you'd do, so you don't have to. Coworkers bombing you on Telegram — Aiko tells you what actually applies to you and replies in your voice. Your mum's birthday — Aiko finds a restaurant nearby and books a table for six at 6:30. *If you can do it on your laptop, ask Aiko.* Under the hood, Aiko is not a wrapper around a frontier LLM. It is a **fractal energy-based harness**: a substrate that routes, verifies, and learns from doing — grounding a small local model with a live index and a shared, auditable memory, so it gets *twice the reliability at half the cost* of running a large model for computer use.

The three pillars (the pitch)

Pillar	The claim	Honest status (see §Status legend)
Reliable	Our own Energy-Based model for computer use, trained on-device; twice the reliability, half the cost	Harness + energy router **BUILT+PROVEN**; the "2× / ½" figures are **DESIGN targets, not yet independently benchmarked**
Private	Weights on-chain; zero-knowledge credential binding; your data stays yours; Solana volume makes the product pay for itself	Local inference + sealed data + reward economics **BUILT/DESIGNED**; **ZK credential binding is RESEARCH-DIRECTION, not built**
Powerful	A new architecture lets Aiko learn *as it works*	Train-on-fire + energy-ranked resolution **BUILT+PROVEN** as a router; "reasoning-grade intelligence" is a **falsifiable bet, not a result**

We state the limits as loudly as the vision — that is the substrate's own tradition, and this dossier keeps it. Where something is a target or a direction, it says so, in the same breath as the claim.

The bet (why this is a different scaling law)

Frontier AI scales intelligence as **parameters × data × compute** — static, centralized, and *your data trains their model*. Aiko scales it as **index coverage × collective firing history × routing/verification quality** — live, distributed, and *your data stays yours*. Knowledge lives in a retrievable index, not in frozen weights, so **the marginal cost of a thought falls toward the cost of a lookup and a fold**. That is the whole thesis; everything else is mechanism.

How to read this dossier

#	Document	What it answers
00	**README-DUE-DILIGENCE** (this)	What is it, what's the pitch, how to navigate
01	**PRODUCT-aiko**	What does the user get; the use cases; why it beats a raw LLM for computer use
02	**ARCHITECTURE-fractal-ebm-harness**	How it works, grounded in the actual code
—	**THESIS-collective-nonparametric-intelligence**	The intellectual bet + the honest wall (foundation)
—	**ECONOMICS-cooperative-commons**	The token, the commons, the reward economics
03	**ONCHAIN-privacy-solana**	Weights on-chain, the frecency cache, the Solana moat, privacy & ZK
04	**PROOF-traction**	What is built and *proven*, with test evidence
05	**RISK-honest-wall**	The risk register — where this breaks and what must be true
06	**DUE-DILIGENCE-FAQ**	Direct answers to the diligence questions

Start at 01 for the product, 02 for the tech, 04 for the proof, 05 for the risks.

Status legend

Every substantive claim in this dossier is tagged:

- **BUILT+PROVEN** — code exists on the `arc-lean` branch with tests that pass in the build gate; verifiable by reading the source and running `zig build test`.
- **DESIGNED** — specified in detail, mechanism worked out, not yet fully built or not yet exercised end-to-end on production data.
- **RESEARCH-DIRECTION** — a bet or an open problem; plausible, motivated, but unproven at this scale. This is where the honest wall lives.

If a claim is not tagged, treat it as narrative framing, not a technical assertion.

Where the company is, honestly

One founder, deep in a two-to-three-month build. The harness is real and running locally; it has one-shotted tasks most agents cannot (see 04). It is **not yet publicly released** — deliberately: the energy-based design is power-hungry and is being contained, and the on-chain layer is being hardened. The pivot from the original Unbrowse (a commons for agents to browse the internet) to Aiko (an agent that "unbrowses" your *own computer's context*) is recent and partial; Unbrowse still exists and is the sister project this inverts. This dossier documents the Aiko direction as it stands today, with the seams visible.

01 · Product — Aiko, the agent on your laptop

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> *Confidential DD dossier. Status tags:* ***BUILT+PROVEN / DESIGNED / RESEARCH-DIRECTION*** *(see* `00-README-DUE-DILIGENCE.md`*).* ***Value/economics layer:*** *how the network funds and rewards itself is the* ***reward-not-slash cooperative commons*** *— usage fees -> USDC to used + verified maintainers, nothing seized —* ***BUILT+PROVEN*** *on* `arc-lean`*. Full model:* `ECONOMICS-cooperative-commons.md` *(and* `03-ONCHAIN-*`*,* `04-PROOF-*`*).*

The product in one line

If you can do it on your laptop, ask Aiko. Aiko is a local agent that does the things you'd do — reading your messages, understanding your context, driving your applications — so you don't have to.

Who it's for

The core customer is **anyone who wants a personal agent that truly understands their context** — the existing user of ChatGPT / Claude / Claude Code who has tried an open agentic tool ("open-*") and found it unusable, unreliable, or unsafe on real work. Aiko is a **better open agent for computer use**: same ambition, but reliable enough and private enough to actually run on your machine, over your real data.

What it feels like (the use cases)

- **Inbox triage in your voice.** Coworkers are bombing you on Telegram. Aiko reads the thread, tells you *what actually applies to you*, and drafts a reply *in your voice* — grounded in how you've replied before.
- **Real-world tasks, end to end.** It's your mum's birthday. "Find a restaurant nearby and book a table for six at 6:30." Aiko finds it, checks availability, and books it — a multi-step task across apps and the web.
- **Anything on the laptop.** The product surface is not a chat box; it is *your computer*. Aiko calls the right binaries and drives the right apps at the right time — less like a chatbot, more like reflexes: a muscle that contracts and fires the correct action for the moment.

Why it beats a raw frontier LLM for computer use

A frontier LLM driving a computer is a brilliant generalist with no memory of what worked here before and no structure forcing it to be rigorous. Aiko wraps the model in a **harness that forces first-principles decomposition**: every task is broken into atomic sub-problems, and the model is made to reason mechanically — *which layer owns this step, where does the problem actually live, what fixes it* — rather than free-associating. The harness is the substrate; the model is one component inside it.

The proof point (BUILT+PROVEN, demonstrated locally): the harness — running Opus as the reasoner inside it — **rebuilt a Wine compatibility layer to launch and run Overwatch on a Mac in effectively one shot.** Not *playing* the game — getting a competitive Windows title to launch on macOS from a standing start. Most agents cannot do this; you cannot simply tell a general agent "make Overwatch run on my Mac" and have it work. It worked because the harness forced the model to solve the routing/travel problem one layer at a time, from first principles, and to locate and fix the actual failure — not because the base model got luckier. (See `04-PROOF-traction.md`.)

The two halves: harness + a small grounded model

Aiko has two parts, and the distinction matters for diligence:

- **The harness is the Energy-Based Model.** It is a *recommendation/routing* system that learns which action to fire next from the collective history of what resolved before (see `02-ARCHITECTURE`). This is the durable, proprietary asset — **BUILT+PROVEN** as a router.
- **A small, non-parametric local model, grounded by the harness.** Separately, a compact instruction-following/reasoning model is being trained to be as *small* as possible so it runs locally. It "doesn't know much" on its own — but because it is **grounded** by the harness (which can feed it live, retrieved

knowledge about how the world and the machine work), it punches far above its size: smarter than the average model of the same footprint. Grounding, not parameter count, is where the capability comes from. **Status: the small model is DESIGNED / in-training; the grounding mechanism (retrieval + energy routing) is BUILT+PROVEN.**

The headline pitch — *"twice the reliability, half the cost"* — is the **design target** of this combination (a small grounded model routed by a cheap energy harness). It is not yet an independently benchmarked result; see the honest wall in `05-RISK-honest-wall.md`.

What "learn as it works" means

Aiko learns **on-fire**: every time it declares an intent, satisfies it, or fails, that verdict folds into the harness's prior immediately — no training run, no labels, no pipeline. Satisfied = good, died = retry. The system's own experience *is* the training signal, and it is realtime: a new fact indexed or a new verdict fired sharpens the next decision now, not after the next retrain. **BUILT+PROVEN** (train-on-fire fold; energy-ranked resolution).

Product status, plainly

Aiko is **used by the founder daily but not yet publicly released.** Two reasons, both honest: (1) the energy-based design is power-hungry and is being contained so it's safe and cheap for others to run; (2) the on-chain layer (which is how the weights become shareable and the storage cost is offloaded) is being hardened. The capability is real and demonstrated; the packaging is in progress.

02 · Architecture — the fractal EBM energy harness

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> *Confidential DD dossier. Status tags:* ***BUILT+PROVEN / DESIGNED / RESEARCH-DIRECTION****. Everything below is grounded in the* `arc-lean` *source tree; file names are real and readable. Foundation:* `THESIS-collective-nonparametric-intelligence.md`*.*

The shape of the thing

Aiko is a single native binary (Zig, zero external dependencies) that is at once: a **contract substrate** (an append-only ledger of intents and verdicts), an **energy-based harness** (a non-parametric router over that ledger), and a **client** to a live index of the world (Unbrowse) and to Solana. There is no Python, no model server, no data files: the embedding is baked in, the "weights" are reconstructible from the ledger, and the reasoning model is grounded through the harness rather than baked into it.

1. The atom — a contract neuron

The unit of everything is a **contract neuron**: one append-only JSONL row recording an intent (a *declare*) and its lifecycle (satisfied / died / purged — the terminal alphabet is exactly those three). A neuron carries its action, its lineage signature (every row is signed from a single wallet-derived root key — `lineage.zig`), and its pointer type. Contracts compose into a graph via `parent_id` containment and `contract:`-ref cross-edges. **BUILT+PROVEN** (`contract.zig`, `ledger.zig`, `pointer.zig`, `terminal.zig`).

The ledger is the shared, auditable memory. It is not a log *of* the system — it *is* the system's state, and the model is reconstructed from it.

2. The energy model — non-parametric, ~1 KB, trained on-fire

The harness's model is deliberately **not** a big parametric net. It is two thin, cheap primitives:

- **Embedding (**`embed.zig`**)** — a deterministic feature-hashing embedding (the hashing trick): each token and character-trigram hashes into a fixed-dimension, L2-normalized vector. Baked into the binary — zero data files, zero network. Cosine over these vectors captures lexical + sub-word overlap. **BUILT+PROVEN.**
- **Energy prototype (**`ebm.zig`**)** — a `Prototype` is a **reward-weighted running-mean centroid** over context vectors (West 1979; O(1) per fire). `foldText(action, reward)` folds a firing in: positive reward pulls the centroid toward it (Hebbian potentiation), negative pushes away (inhibition — the contrastive negative phase). The **energy** of a context is its negative cosine to the prototype; routing is **argmin energy**. **BUILT+PROVEN.**

Each such head is ~1 KB and reconstructible from the ledger. **The knowledge is not in the weights** — it is in the retrievable index (the world) plus the collective firing history (the ledger). A thought costs a retrieval + a centroid fold, not a forward pass through a giant net.

3. Fractal keys — multi-scale priors for free

The same firing folds into the *same atom* at many scales at once — `invoke:{event}`, `:L{layer}`, `:dim:{w6h}`, `:verb:`, `:ptype:`, `:root`. One cheap operation yields a **coarse-to-fine router**: the collective can prioritize at every granularity — per-event, per-layer, per-dimension — **without a separate model per scale.** This is the "fractal" in the name: the EBM is not one model but many nested thin ones, each wrapping a different facet of a route, broken down to first principles. **BUILT/DESIGNED** (fractal key convention in `fractal.zig` / `pattern.zig`; the arc-lean router exercises the per-run centroid directly).

4. Energy-ranked resolution — the router IS the harness

This is the load-bearing mechanism and the falsifiable atom of the whole thesis. When the substrate has a frontier of candidate contracts to fire next, it does **not** scan them in arbitrary order. `energyRankCandidates` (in `main.zig`) reconstructs a `Prototype` from the run's resolved history (each satisfied action folded with reward +1, each died with −1 — the train-on-fire signal, the ledger *is* the weights), then **sorts the frontier best-first by energy** so the most "resolved-like" candidates fire earliest — shorter cascades, less wasted work.

It is **always on** and **reorder-only**: every candidate still fires, so nothing is starved (a fairness floor holds by construction). It is the **thinking-budget allocator** — it spends scarce compute/spend where history says it pays. **BUILT+PROVEN**, and *measured*: on the prove-or-kill benchmark it settles a producer/consumer graph in **2 passes vs 3** for scan-order, mutation-verified (reversing the sort restores 3 — so the ranking, not luck, is doing the work). This is the atom "the collective routes its own thinking," proven before the cathedral. (See `04-PROOF-traction.md`, `PLAN-contracts-as-onchain-ebm-weights.md`.)

5. Ledger-reconstructible weights — the model is the commons

Because every weight is a fold of ledger rows, **anyone can rebuild the prior from the shared ledger** (`replayFromLedger`, on `main`). The "weights" are the collective experience, in the open — forkable, inspectable, re-derivable. No single node is "the model": the graph of contracts + the ledger-reconstructed EBM

- the indexed world is the mind. This is also what makes the weights **portable on-chain**: a weight is just content-addressed ledger data (see `03-ONCHAIN-privacy-solana.md`, and `cas.zig` — content-addressed store, dedup-for-free, **BUILT+PROVEN**).

6. Grounding a small model — capability without parameters

The reasoning model is trained to be **as small as possible** (non-parametric in spirit: it holds instruction-following and reasoning, not world-knowledge). On its own it "knows little." But the harness **grounds** it: retrieval brings the relevant slice of the world and the machine to bear, and energy routing tells it which action history says works. A small model grounded this way is **smarter than the average model of the same footprint** — capability comes from grounding, not scale. **Status: the small model is DESIGNED / in-training; the grounding mechanism (retrieval + `embed`/`ebm` energy routing) is BUILT+PROVEN.** The leap from "thin router" to "reasoning-grade intelligence" is the honest open problem — stated plainly in the THESIS and `05-RISK-honest-wall.md`.

7. The module map (what's real, by concern)

Concern	Modules (`libcontract/src/`)	Status
Energy harness	`embed`, `bge` (baked BERT), `ebm`, `route` (`recommend`), `pattern`, `fractal`	**BUILT+PROVEN** (router measured — 88.9% LOO, `route_bench.zig`)
Contract substrate	`contract`, `ledger`, `pointer`, `terminal`, `cache`, `guards`, `capability`	**BUILT+PROVEN**
Identity / truth-root	`lineage`, `keystore`, `bible`, `verses`	**BUILT+PROVEN**
Commons economics	`cas`, `reward`, `usage`, `usage_ledger`, `epoch`	**BUILT+PROVEN** (local)
Solana / on-chain	`voltr`, `stake`, `payout`, `broadcast`, `broadcast_card`, `solana_rpc`, `solana_tx`, `base58`	**BUILT+PROVEN** (read/derive live; write = operator-gated)
Small reasoning model	(separate training track)	**DESIGNED / in-training**

Every module ships its tests *inside its source file*, exercised by `zig build test` and gated by `scripts/build.sh` (unit tests + 53 doctrine-parity claims + a signed self-attestation). Green build = every claim above has matching, tested code.

03 · On-chain, privacy & the Solana moat

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> *Confidential DD dossier. Status tags:* ***BUILT+PROVEN / DESIGNED / RESEARCH-DIRECTION****.*
> *Companion to* `ECONOMICS-cooperative-commons.md` *(the token + reward economics).*

Why on-chain at all

Aiko's energy-based design has one honest cost: **it grows storage as it learns.** Every action trains a prototype and folds new history; run it long enough on one laptop and the weights + ledger accrue. The design turns that liability into a feature by **putting the weights on-chain**:

- **Weights become shareable.** A weight is just content-addressed ledger data (`cas.zig` — content-addressed store, dedup-for-free, integrity-checked; **BUILT+PROVEN**). Anyone with access can reconstruct the same prior from the same rows. You can *call your AI* — hand it to another device, fork it, audit it.
- **Storage cost is offloaded.** The machine holds a **frecency cache** — hot, frequently/recently used weights persist locally; cold ones live on-chain and are pulled on demand. Store **primarily on-chain**, cache locally what's in frequent use. **DESIGNED** (the caching policy; `cache.zig` is the local layer; the on-chain tier uses the anchoring path below).
- **Shared data stays cheap.** Content-addressing means a weight referenced by many contracts is stored (and paid for) **once** — the dedup is a property of the addressing, not a special case. **BUILT+PROVEN** (`cas.zig`).

Status of full on-chain weight storage: DESIGNED, partially proven. The primitives are real — content addressing (`cas`), live Solana reads (`solana_rpc`, devnet-proven), on-chain hash anchoring (`iq_ledger` on `main`, anchors claim hashes to a deployed Solana program), and a rent model that bounds what goes on-chain (≤ ~4 KB on-chain, larger = commitment-only + off-chain). The end-to-end "weights live on-chain, laptop is a cache" loop is assembled from these but not yet running in production at scale.

The Solana moat

The strategic bet is distribution + storage, stated plainly:

> *Once the EBM approach is shown to work, frontier labs will copy it — it's proprietary tech, but it's copyable.* ***The moat is not the algorithm; it's the ledger.*** *No one in AI has a distributed ledger the size of Solana. A ledger that can hold all the weights in the world — and a distribution channel already in millions of wallets — is the durable asset. Win the distribution and the storage substrate before the technology is commoditized.*

And the economics close on themselves: **increased Solana transaction volume means the product pays for itself.** Every weight write, every attestation, every settlement is on-chain activity; as usage grows, the on-chain volume it generates funds the substrate it runs on. (The full reward economics — usage fees paid out to stakers in USDC, FDRY as the security stake, reward-not-slash — are in `ECONOMICS-cooperative-commons.md`, **BUILT+PROVEN locally**, with the real mainnet FDRY vault integration in `voltr.zig`, **BUILT+PROVEN**.)

Honest status of the moat: this is a *go-to-market and defensibility thesis*, not a technical lock-in that exists today. Solana gives distribution + cheap durable storage; it does not by itself prevent a well-funded lab from building the same harness on their own infrastructure. The moat compounds only if Aiko wins the network (the commons of shared, corroborated weights) first. See `05-RISK-honest-wall.md`.

Privacy — what holds, and what is a bet

The pillar is *"your information stays yours,"* and it decomposes into three distinct claims with three different maturities. Diligence should not conflate them:

- **Local inference + sealed data — BUILT/DESIGNED.** Inference runs *on your machine* over your own data plus a shared index. You don't ship your data to train someone else's model; you fire locally and contribute **signed, capability-gated verdicts** — attestations of *what worked*, not the content. Your data at rest is

sealed under your key (envelope encryption + the **receptor grant** access boundary — `capability.zig`'s opt-in receptor is **BUILT+PROVEN**: value is adopted only under a signed grant, measured at 100% cross-lineage enforcement).

- **Zero-knowledge credential binding — RESEARCH-DIRECTION (not built).** The pitch says "ZK-proof credential binding creates privacy you can verify." Be precise for diligence: **there is no production zero-knowledge system today.** The current on-chain identity/attestation layer uses hash commitments and signed lineage, *not* a ZK proof system. Real ZK credential binding (e.g. Pedersen/Ristretto commitments + Schnorr + KVAC, avoiding SNARK trusted setup) is designed on paper and is the intended path — but it is unbuilt. Treat "ZK" as a **direction**, not a shipped feature.
- **Query privacy over a shared index — UNSOLVED.** Your data *at rest* is private. But *querying* a shared index can leak what you're asking. True query privacy needs PIR / local replicas / ZK — a separate, hard problem, not solved by encryption-at-rest. **"Private" holds for your data; not yet for your attention.** (Stated identically in the foundation THESIS — this is not new caution, it's the standing honest wall.)

The one-line summary for diligence

On-chain is **real in its primitives** (content-addressing, live reads, anchoring, a real FDRY vault, the reward loop) and **a designed system** at the "weights fully live on-chain, laptop is a frecency cache" level. Privacy is **real for data-at-rest**, **a direction for ZK**, and **an open problem for query privacy**. The moat is a **distribution/storage thesis**, compelling but not a technical lock-in yet. Everything here is labelled so the seams are visible.

04 · Proof & traction — what is built and proven

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The build gate (the honesty mechanism)

The whole substrate ships behind one gate: `scripts/build.sh` runs the unit test suite, **53 doctrine-parity claims** (mechanical checks that the code matches its own stated doctrine), a formatting pass, and a **wallet-signed self-attestation** of the binary. A green gate means every module's in-source tests pass *and* the code has not drifted from its documented invariants. This dossier's **PROVEN** tags are all downstream of that gate. It is currently green: *53/53 claims, signed.*

The falsifiable atom — energy-ranked resolution beats scan-order

The single most important proof, because the entire thesis rests on it: **does the collective's energy prior actually route thinking better than arbitrary order?**

- **Result: yes, measured.** On a prove-or-kill benchmark (producer/consumer contract graph), energy-ranked resolution settles the graph in **2 passes vs 3** for scan-order — fewer cascades, less wasted work — **16/16** runs.
- **Mutation-verified:** reversing the energy sort restores 3 passes. So the *ranking*, not luck or an unrelated change, is doing the work.
- **Non-breaking:** it is reorder-only; every candidate still fires (fairness floor holds). It is now **always on** — the default resolution path.

This is "the collective routes its own thinking," proven at the atom before building the cathedral. **PROVEN** (`main.zig` router; `examples/arc-agi3/ bench-energy-router.sh`).

The product one-shot — Wine/Overwatch on macOS

The harness (with Opus as the reasoner inside it) **rebuilt a Wine compatibility layer to launch and run Overwatch on a Mac, in effectively one shot** — a competitive Windows title brought up on macOS from a standing start. This is the capability proof: most agents cannot do this, and you cannot get there by telling a general agent "make Overwatch run on my Mac." It worked because the harness forced first-principles, layer-by-layer decomposition and fault localization. **Founder-demonstrated** (a Loom/recording is the artifact to request for diligence).

ARC-AGI benchmark — reasoning without a large model

The founder tested an earlier version against **ARC-AGI-3** (a benchmark designed so that passing normally requires a large model) and **reached a high score without using any large model** on the order of Grok/Opus — the point being that the *harness*, not raw parameter count, carried the reasoning. **Founder-reported; awaiting independent replication** (the arc-lean tree is mid-rebuild toward that earlier capability — `examples/arc-agi3/`). Diligence should request the exact score, the task split, and a reproducible harness invocation.

The commons economics — proven locally, end to end

The cooperative-commons economics (`ECONOMICS-cooperative-commons.md`) are not a whitepaper sketch; the mechanism runs and is tested:

- **Reward, not slash.** Usage fees form a pool paid out to the maintainers whose routes were *used* and still *verify* — usage-weighted, verification-gated, stake-gated. Tested invariants: **conservation** (nothing burned), **forfeit -> value** (a route that fails re-verification forfeits its fees to the honest, who then earn more than their own usage), **stake-gated** (unstaked earns nothing). **PROVEN** (`reward.zig`).

- **Wash-trading is a donation, not a theft.** Sybil-dampened usage accounting (self-exclusion + per-payer cap): a maintainer wash-calling its own route earns **0** and the honest take the pool. **PROVEN** (`usage.zig`, `usage_ledger.zig`).
- **The whole loop runs from the real ledger.** `epoch.runEpoch` + `runEpochFromLedger` assemble stake (from chain), owner + attestation + usage (from ledger rows) and produce the payout set — a capstone test drives every gate at once from JSONL. **PROVEN.**

Real chain integration — reads live, writes are operator-gated

- **FDRY staking is real.** `voltr.zig` builds native Voltr vault instructions against the **real mainnet vault**; its PDA derivation **reproduces the actual on-chain vault authority** from seeds — a wrong byte fails the build. Cross- validated against Voltr's published CPI reference (program id, discriminators, account order, seeds all match). **PROVEN, without spending anything.**
- **Live reads work.** `solana_rpc` fetches live devnet blockhashes and reads a wallet's stFDRY stake from mainnet (`stake.zig`). **PROVEN.**
- **The submit path is cluster-valid.** `broadcast.zig` compiles + signs a legacy transaction; submitting it to devnet with a dummy blockhash returns `BlockhashNotFound` — i.e. the wire **deserialized and reached simulation** (a malformed wire errors "failed to deserialize"). So the serialization is byte-correct against the real Solana runtime. The operator utility `broadcast-tool` runs the full pipeline end-to-end (load key -> parse card -> sign -> submit). **PROVEN.**
- **Money moves only by the human's hand.** Every value-moving step emits a *card* and never sends; only the operator, running `broadcast-tool` with a funded key, fires it. This is a deliberate safety boundary, not a missing feature.

What "traction" honestly means today

- **Working software, daily use by the founder.** The harness is real and running.
- **Zero external users / not publicly released.** By choice (energy containment + on-chain hardening), not for lack of a build. This is pre-launch.
- **Reproducible technical proofs** (the router, the commons loop, the chain integration) are in-tree and gated.
- **Capability proofs** (Overwatch one-shot, ARC-AGI score) are founder- demonstrated and are the right things to ask for a recorded, reproducible demo of. The honest ask for diligence: a Loom of the harness one-shotting a fresh task, and the ARC-AGI run with its numbers.

05 · Risk register — the honest wall

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> *Confidential DD dossier. The substrate's own tradition: state the limits as loudly as the vision, or it's marketing. This is the section diligence should weight most. Nothing here is hidden elsewhere; it is collected for scrutiny.*

Technical risk

R1 — The energy model is a router, not a mind (the core bet). Today's model is a cosine-to-centroid plus a thin linear head — soft priors over "what resolved before." Real reasoning is **not** in this model. The leap from "thin router" to "intelligence" is *unbuilt* and comes from one of two unproven places: richer composed energy models (which return some of the compute cost) or emergent collective composition (many thin routers + verification composing into reasoning). **This is the whole company's central technical bet.** Mitigation: the falsifiable-atom discipline — prove each rung (the router *is* proven) before funding the next. Do not mistake the scaffold for the building.

R2 — Retrieval ≠ reasoning. Indexing the world gives knowledge *access*, not thought. Superintelligence needs composition, planning, and verification *over* retrieved knowledge. The energy-descent-as-generation primitive is the seed, but it is linear and shallow. **The reasoning layer is the hard, unsolved core.**

R3 — Verification is the crux and the single biggest risk. A system that learns on-fire from verdicts is only as good as its verdicts. Bad or *gamed* verdicts don't just add noise — they train the shared prior toward delusion. The guard (a canonical, fixed reference anchor + independent multi-party corroboration + the reward economics' verification gate) is **load-bearing**: if it fails, the collective becomes confidently, collectively wrong. This is the line between wisdom-of-crowds and madness-of-crowds, and it is **not automatic.**

R4 — "Low compute" is low *marginal*, not low *total*. Embedding and serving retrieval over the live web is heavy infrastructure. The compute *moves* from training to indexing/serving; it does not vanish. Honest claim: cheap-per-thought, not cheap-overall. The "half the cost" pitch is a per-task target, not a total- infrastructure claim, and is **not yet independently benchmarked.**

R5 — Energy containment (the founder's own stated blocker). The design is power-hungry — it "leaks energy" by nature (it trains and grows storage on every fire). This is *why* it isn't released yet. The on-chain offload is the intended fix (weights on-chain, laptop as frecency cache), and is DESIGNED but not proven at scale. Until contained, the product is unsafe/uneconomical to hand to users.

R6 — Query privacy is unsolved; ZK is unbuilt. Data-at-rest is sealed, but querying a shared index can leak intent (needs PIR/local-replicas/ZK). And the headline "ZK credential binding" is a **direction, not a shipped feature** — today's identity layer is hash commitments + signed lineage. Marketing that says "privacy you can verify" is ahead of the code; diligence should hold the code's actual maturity (see `03-ONCHAIN-privacy-solana.md`).

Strategic / market risk

R7 — The algorithm is copyable; the moat is the network, not the code. If the EBM harness works, frontier labs can and will build it — it is proprietary but not un-reproducible. The defensibility thesis is **distribution + the ledger** (Solana's storage + wallet distribution + a commons of corroborated weights won first). That is a race, not a lock. If Aiko does not win the network before the approach is commoditized, the moat does not hold.

R8 — Single-user, not yet a collective. The thesis's power (compounding intelligence, shared prior) requires *many* agents firing and corroborating. Today there is **one user** (the founder). Every "collective" property is architectural potential, not observed behavior. The network effects are unproven because the network does not yet exist.

R9 — Pre-launch, zero external validation. Not publicly released, no external users, no revenue. Capability is founder-demonstrated (Overwatch one-shot, ARC-AGI) and awaits independent, recorded replication.

R10 — A new-currency ambition adds scope and regulatory surface. The founder's end-vision includes defining a *new currency that rewards novelty/value-created* (beyond value-of-work). This is early and unformed, and paying token holders from usage introduces securities-like questions. It should be treated as a research

aspiration, not a near-term deliverable, and de-scoped from the core investment thesis.

Execution / key-person risk

R11 — Single founder, deep in a rabbit hole. One person, ~2–3 months into a very deep build, self-aware about scope creep ("dug myself into a deep rabbit hole"). Enormous surface area (harness + model + on-chain + economics) for one person. Key-person risk is maximal; the first hire and a scoped roadmap are the obvious de-risking moves.

R12 — Two projects in superposition. Aiko is a partial pivot from Unbrowse; Unbrowse still exists. Focus risk: the pivot is not clean, and "which company is this" is a real question a first-round panel already flagged.

"Superintelligence" — a bet, not a result

Collective systems (markets, evolution, science) *do* exceed any participant, so the ambition is not crazy. But whether this reaches *super* — and not merely *useful* — is untested and hinges entirely on R1, R2, and R3 holding. It should be underwritten as a **falsifiable direction with a cheap first gate already passed** (the router), not as a destination being claimed.

The one honest sentence

The router is proven; the reasoning is a bet; the network doesn't exist yet; the privacy pillar is real for data and aspirational for ZK; and the moat is a race to own a commons before the idea is copied. Everything compelling about Aiko is downstream of bets that are named, cheap to test next, and being tested in order.

06 · Due-diligence FAQ

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> *Confidential DD dossier. Direct answers to the questions raised in the second-round diligence conversation. Cross-references point to the fuller treatment. Honest status tags throughout.*

Q1 · What are you actually building? What's the core product and problem?

Product: Aiko — a local agent that does the things you'd do on your laptop, so you don't have to (triage your Telegram in your voice; find a restaurant and book the table; *if you can do it on your laptop, ask Aiko*). **Problem:** existing agentic tools for computer use are unreliable, expensive, and privacy-hostile — brilliant generalists with no memory of what worked here and no structure forcing rigor. **Core insight:** instead of "unbrowsing" the *internet* (the original Unbrowse), Aiko "unbrowses" the *context of your own computer* — building reflexes that call the right binary at the right time, from an agent that actually understands you. See `01-PRODUCT-aiko.md`.

Q2 · Who is the core customer?

Anyone who wants a **personal agent that truly understands their context** — the current ChatGPT/Claude/Claude-Code user who tried an open agentic tool and found it unusable. Aiko is a **better open agent for computer use.** See `01`.

Q3 · How is this different from what frontier labs ship?

Frontier labs are **parametric-LLM-centric** — intelligence ∝ params × data × compute, static, and *your data trains their model*. Aiko is **harness-centric**: intelligence ∝ index-coverage × collective-history × routing/verification, live and local, and *your data stays yours*. Very few players take **energy-based models** seriously because the naive version needs a data-center to "simulate reality." The breakthrough is **breaking the EBM into fractal atoms** — not one giant EBM, but many tiny ones, each wrapping a route from first principles — which makes it cheap and local. See `02-ARCHITECTURE`, foundation THESIS.

Q4 · Are people using it today?

The **founder uses it daily; it is not publicly released.** Two honest reasons: the energy-based design is power-hungry and is being contained (R5), and the on-chain layer is being hardened. Zero external users today — pre-launch by choice, not for lack of a working build. See `04-PROOF-traction.md`, `05-RISK`.

Q5 · When did you pivot from the browser approach?

Partial, recent pivot (~2–3 months), not complete. Unbrowse originally built a *commons for agents to browse the internet*. As agents/AI improved, they learned to browse anyway — diminishing returns on a route-commons. Inverting the idea — point the same route-engineering at *your computer* instead of the web — is Aiko, started for the Colosseum hackathon. Unbrowse still exists; the two are sister projects (R12). See `00-README`.

Q6 · If the EBM approach works, why wouldn't OpenAI/Anthropic copy it?

They would — it's copyable. The moat is **not the algorithm; it's the ledger + distribution.** No one in AI has a distributed ledger the size of Solana to hold all the weights, plus wallet-level distribution. Win the storage substrate and the network of corroborated weights before the technique is commoditized. **Honest:** this is a *race*, not a technical lock (R7). See `03-ONCHAIN`.

Q7 · How close are you to containing the energy usage?

Founder estimate: **~95% — the remaining work is stability, not invention.** In principle it can run fully on-chain and even remotely (not on your machine), because the design lets you push data on-chain by paying some SOL. **Honest status:** the on-chain offload is **DESIGNED and partially proven** (real reads, real vault, anchoring primitives) but the full "weights on-chain, laptop is a frecency cache" loop is not yet running at scale (R5). See `03`.

Q8 · How do you measure whether it's working?

The falsifiable atom: **does energy-ranked resolution beat scan-order at choosing what fires next?** — **yes, measured: 2 passes vs 3, mutation-verified** (`04-PROOF`). At the capability level, **ARC-AGI-3**: an earlier version reached a high score **without any large model** — the harness carried the reasoning. **Founder-reported, awaiting independent replication** (request the score, split, and a reproducible invocation).

Q9 · The one-shot game build — did you pre-train it? How did it learn?

No task-specific pre-training. At that time it was **harness + Opus** (the model as reasoner *inside* the harness), which **rebuilt a Wine layer to run Overwatch on a Mac in one shot.** It works by forcing the model to think in **first principles** — decompose the route/travel problem layer by layer, locate where the failure actually is, and fix it — made mechanical and instinctive by default, rather than free-associating. See `01`, `04`.

Q10 · Is this more a harness than a model? Is the harness the EBM?

Yes — it's primarily a harness, and the harness *is* the energy-based model. **The EBM is the recommendation/routing system** that learns which action to fire from collective history (BUILT+PROVEN). *Separately*, a **small non-parametric reasoning model** is in training — kept as small as possible so it's local, and made powerful by being **grounded** by the harness (retrieval + energy routing) rather than by parameter count. A small grounded model beats a same-size ungrounded one. See `02-ARCHITECTURE §6`.

Q11 · What's the next step if it works?

Beyond automating enormous amounts of work, the founder's stated end-vision is to **accelerate the creative process** and eventually **define a new currency** — one that rewards *value created / novelty*, not value-of-work. **Honest:** this is an early, unformed aspiration with real regulatory surface (R10) and should be de-scoped from the near-term investment thesis; the near-term is: contain the energy, ship to first users, prove the collective with more than one agent.

Q12 · What can we diligence right now?

- **This dossier** (00–06) — product, architecture grounded in real code, economics, on-chain/privacy, proof, and this FAQ.
- **The code** — the `arc-lean` tree; run `bash scripts/build.sh` (unit tests + 53 parity claims + signed attestation) to verify the **PROVEN** claims yourself.
- **The best next artifacts to request:** a recorded (Loom) demo of the harness one-shotting a fresh computer-use task, and the ARC-AGI-3 run with its exact numbers and a reproducible command.

Bottom line for the panel: the hard, cheap-to-test atom is proven (the router routes better than chance); the grand claim (reasoning-grade collective intelligence) is a named, ordered bet; the moat is a race to own a commons; and every seam is labelled in `05-RISK-honest-wall.md`. This is a research-risk investment with an unusually disciplined honesty culture and a working, demonstrated harness — underwrite the founder and the falsifiable next gate, not the destination.